

Prediction of the Annual Electricity Generation from the UK Offshore Wind Farms by Analysing the Factors Influencing Wind Resource using Machine Learning

Abstract

Offshore wind farms will play a significant role in power supply as the UK moves towards zero-carbon emissions. That is why it is increasingly important to be able to forecast power for the future. The ability to forecast power will be advantageous in improving system operations because it can reduce operating costs, unserved energy and wind curtailment. Moreover, it will also help maintain the required levels of system reliability. It is well established that wind speed is the main factor influencing the power produced by the turbine. The power curves can be used to estimate power output based on wind speed; however, due to the unpredictable nature of wind and relationships with other meteorological factors, this method does not always produce accurate results. The use of machine learning algorithms will help understand these factors' significance and further develop more precise wind power forecasts. This paper analysed the meteorological mast data from the 7 MW Levenmouth Demonstration Turbine in Fife, Scotland. The analysis led to the discovery of multiple anomalies. Anomaly detection techniques and a filter were used to remove anomalies to make the data less noisy. Tree-based and deep learning models were trained on the dataset to make predictions. They were chosen based on their reputations as some of the most powerful and robust learning algorithms for capturing non-linear relationships. The models showed promising results with r-squared values of up to 96% and feature importance scores that showed only wind speed as the most significant factor and other meteorological factors having negligible importance in forecasting wind power.

Keywords: offshore wind farms, machine learning, feature importance, power forecast

1. Introduction and Background

A clean and reliable power system will be the foundation of the UK's Net Zero Strategy, which aims to set out policies and proposals to decarbonise all sectors of the UK economy by 2050 ([UK Government, 2021](#)). Renewable energy from offshore wind farms will be a significant factor. The UK is the world leader in offshore wind, which currently powers over 7.5 million homes. In addition, it is the home of the largest wind farm and has more installed capacity than any other country. As a result, offshore wind is one of the country's most significant growth industries, creating skilled job opportunities for thousands and attracting billions in investments ([Ørsted, n.d.](#)). By 2030 the UK Government expects to deliver 40GW offshore wind power (plus an additional 1GW from floating offshore wind) and 75GW by 2050 ([Ministry of Defence, 2021](#)).

There are multiple techniques for wind power forecasting. The simplest method assumes that power in the next interval will be the same in the previous interval. Unfortunately, this is a very naïve method, and its effectiveness deteriorates with time. Physical methods typically use the turbine power curve, which illustrates the relationship between output power and hub height wind speed ([Sohoni et al., 2016](#)), to make forecasts. That said, the nature of wind power is characterised by strong fluctuations,

uncertainty and intermittency ([Saroha, S., Aggarwal, S. K., Rana, P., 2021](#)), which lessens the capability of this method. Finally, using historical data, statistical methods develop linear and non-linear relationships between weather prediction factors and power. They are advantageous over other methods. In the study by Hanifi et al., it was discovered that physical methods are more complex and resource intensive under the same conditions while producing worse metrics than statistical methods ([2020](#)). Some standard statistical procedures are Tree-based models, Artificial Neural Networks and ARIMA.

Various projects have applied statistical methods to forecast power generated from wind turbines. For example, Clifton et al. used simulated data of a 1.5 MW turbine to train a combination of regression trees (Random Forest). Wind speed, shear and turbulence were used as input for the model and the results were ranked to measure their influence on power generation. The model produced three times better metrics than the turbine curve ([2013](#)).

Lin et al. used SCADA data from a Levenmouth Demonstration Turbine as input for a 5-layer deep learning neural network. Isolation Forests and Elliptical Envelopes were used for detecting outliers. Both methods were trained on a deep learning model, and it was discovered that models trained on datasets cleaned with Isolation Forests performed better than the elliptical envelopes ([2020](#)).

Khan et al. used a hybrid transfer learning approach that consisted of two spaces; The first from an already trained model. The second was a small feature set gotten from a windfarm used for retraining. After training, the model had normalised RMSE of as low as 0.03 and had accuracy losses that were less than 5% ([2022](#)).

Alkesaiberi et al. took multiple approaches, such as univariate forecasting of wind power data to predict the trend of future wind power. In this case, only the past values of wind were considered. The other method made predictions on the meteorological variables as input. Experimentations were done using machine learning models such as Random Forest Regression, Gaussian Process Regression, and Support Vector Regression. The experiment results revealed that there was no single approach dominating. Still, the Gaussian Process and Tree models had the best metrics, with R-squared values of about 0.95 in different experiments ([2022](#)).

2. Aims and Objectives

With all this energy expected to come from wind farms, there is a need to reduce vulnerability to sudden price rises caused by fluctuating international fossil fuel markets. The hope is that the cost of clean energy will be cheaper and more competitive than carbon-laden gas. That is why it is adamant to make accurate forecasts on power generation, as forecasting electricity is one of the most important exercises when managing power systems. Studies have shown that forecasting wind power can improve system operation, maintain system reliability, and reduce operating costs ([Lew et al., 2011](#)).

This project aims to investigate the factors that influence the power generation of offshore wind turbines to aid in making higher precision forecast for wind energy. The following objectives need to be completed:

- Analyse dataset to gain insights on the factors influencing power generation.
- Develop a machine learning model to find patterns in the dataset to make accurate forecasts.
- Extract feature importance to evaluate each factor's significance in the power output.

3. Methodology

3.1. Data Collection

For this research, data from a 7 MW Turbine located in the Methil Offshore Windfarm, a demonstrator site for wind turbines, was obtained from Ore Catapult ([ORE Catapult, n.d.](#)). Two datasets were used for this project:

- **Met Mast:** This is meteorological data from the turbine site gotten from 11 sensors. It contained wind speed readings at four different hub heights, temperature, air pressure and wind direction.
- **Turbine:** This is data gotten using 574 sensors on the turbine itself. It contained electrical readings such as power generated by the turbine.

The data was obtained using Supervisory Control and Data Acquisition (SCADA). They are software applications for gathering real-time data from remote locations to control equipment and conditions ([European Wind Energy Association, 2009](#)). The records of the data were in 10-minute intervals. In addition, each feature's mean, maximum, minimum, standard deviation and end value were presented. The datasets started from 2017 till 2021.

3.2. Analysis

The table provides much insight. Firstly, each count is not up to 259611, meaning there are missing variables. The mean and median of the temperature and power entries significantly differ. This could result from outliers or a skewed distribution as the mean is very sensitive to outliers, unlike the median, which is less so ([Skiena, 2017](#)). The presence of an outlier can be confirmed by the minimum temperature value, which has a value of $-1.57 \times 10^{16}^{\circ}\text{C}$, which is an impossibly low-temperature value. Finally, there are negative values in the power entry, as seen in the 25th percentile column. The power generated by wind turbines should never be negative. This might be a fault in the data collection process. Some sources have said that this could result from the turbine absorbing power instead of producing it ([Younas Zubair, 2019](#)). Regardless this is an error that needs to be fixed.

Table 1 Descriptive statistics of dataset features

	Count	Mean	Median	Std	Min	25%	75%	Max
Power (W)	255872	1224.91	0.00	2014.92	-129.32	-0.75	1676.03	7149.04
Wind Speed (m/s)	244164	7.07	6.73	3.92	0.00	4.01	9.57	30.78
Temperature (°C)	245069	-6.40×10^{10}	11.62	3.17×10^{13}	-1.57×10^{16}	8.13	14.98	26.59
Direction (°)	245069	189.50	222.40	90.41	0.00	101.88	252.56	359.99
Pressure (hPa)	244164	1007.12	1019.33	105.48	0.00	1011.09	1025.53	1048.06

The Wind Rose is a polar plot that displays the frequency of wind by direction ([Brower, 2012](#)). From **fig 1**, the highest proportion of winds come from around the South West, and most wind speeds range from 6.2-12.3 m/s.

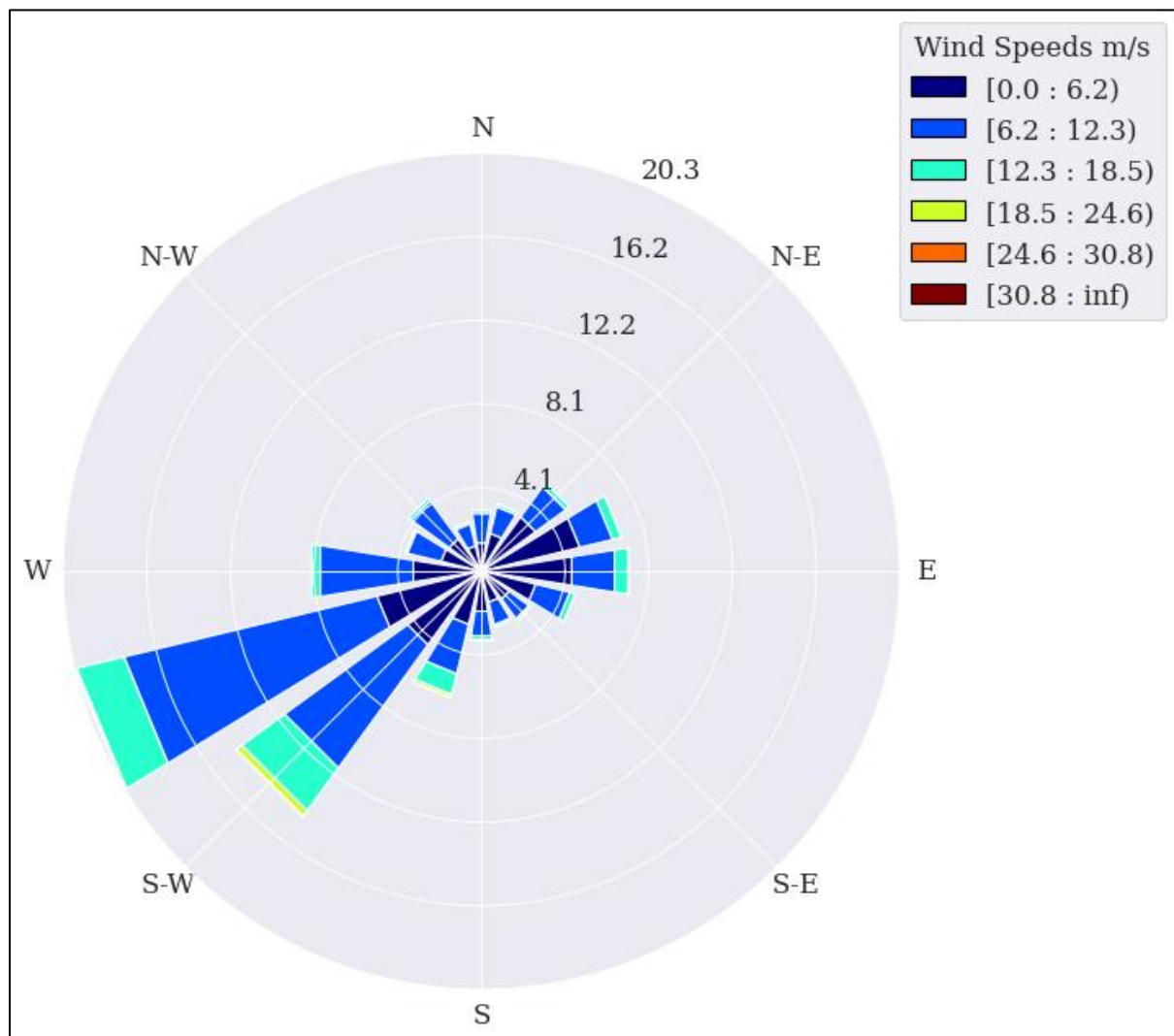


Fig 1 Wind Rose plot

Fig 2 shows how wind speed is distributed in bins of 1m/s. The Weibull distribution is a function that represents the wind speed frequency. It is given by the equation:

$$p(v) = \frac{k}{A} \left(\frac{v}{A}\right)^{k-1} e^{-\left(\frac{v}{A}\right)^k}$$

The value A is the scale parameter, closely related to the mean wind speed value, and its unit is in m/s. The value k is the shape which is non-dimensional. It determines the width of the distribution. The value of k is commonly observed around 2. The shape parameter around this level is called a Rayleigh Distribution ([Guthrie, 2020](#)). The value v represents the wind speed.

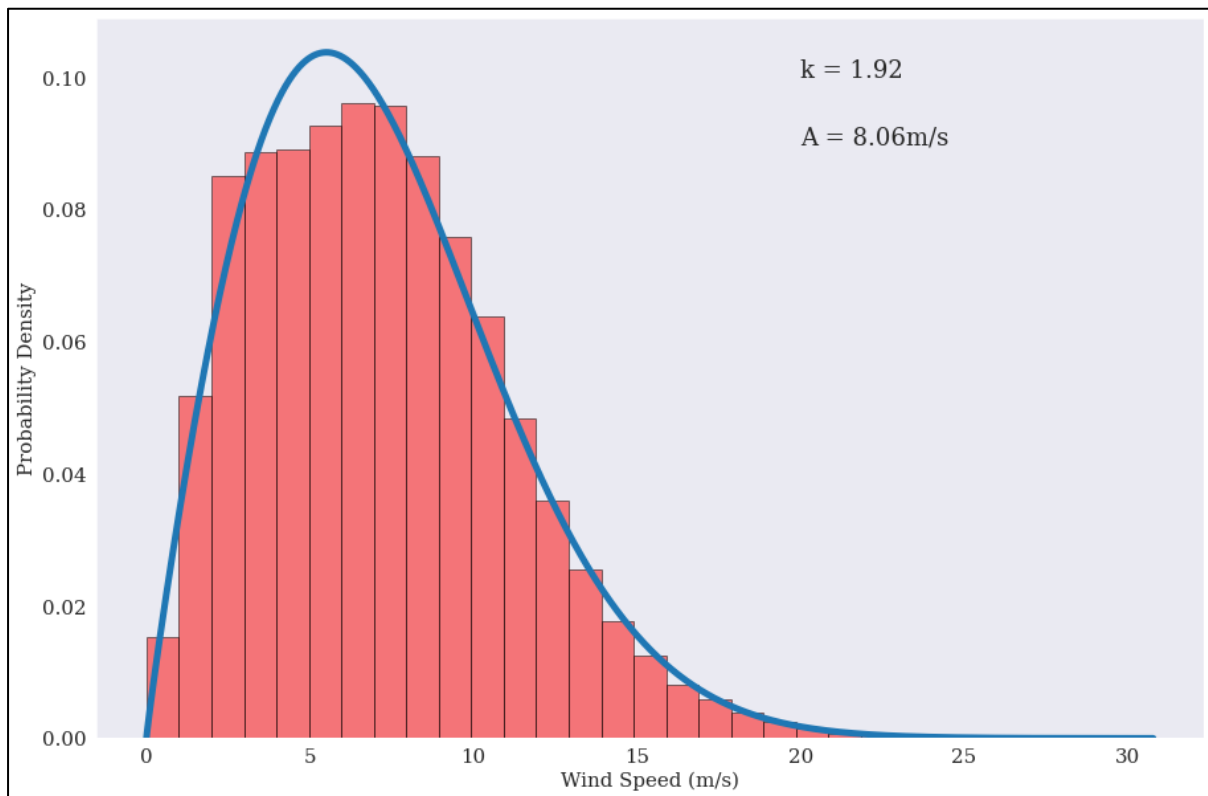


Fig 2 Wind Speed distribution with best fit Weibull curve.

An autocorrelation plot is used in time series data to show whether the elements are correlated. In other words, it shows if the entries in previous time steps are directly related to the entries occurring after it ([Anderson et al., 2015](#)). The values for the autocorrelation function range from 1 to -1. When values are closer to 1, they are positively correlated, while closer to -1 means negative correlation. If there is no correlation, the value should be near zero. The x-axis represents the lags which show how many time steps ahead we are looking at for autocorrelation. As shown in **fig 3**, there is high autocorrelation for all the factors of the dataset, meaning previous values are directly related to the one in the next time step.

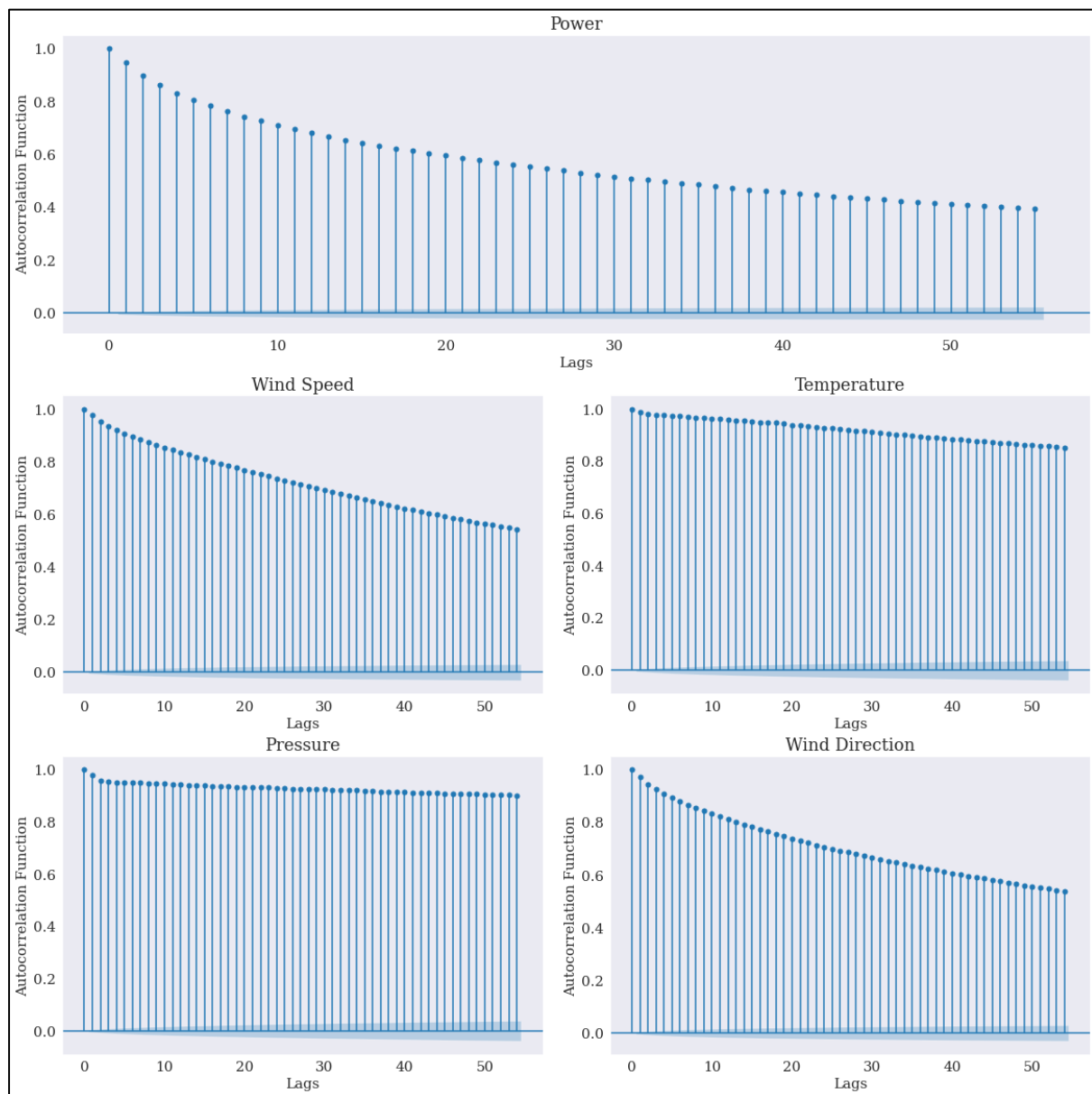


Fig 3 Autocorrelation Plot for each feature in dataset.

The power curve depicts the relationship between a turbine's power output and wind speed. It aids in displaying the turbine performance [\(Lydia et al., 2014\)](#). The power curve manufacturers produce typically has a sigmoid shape; however, the turbine results usually deviate from this ideal shape depending on the environment. The power curve also displays the turbine's rated, cut-in and cut-off wind speeds. In **fig 4** below, the black line shows what an ideal curve might look, and the ones in blue are the actual points. We see the points follow the shape of the ideal curve, although some points deviate so far out they can be considered anomalies. Anomalies are instances that do not follow the pattern of the rest of the data. Removing these anomalies is essential so they do not bias the data. There are usually three categories of anomalies in wind turbines classified by [\(Morrison et al., 2022\)](#). The different types of anomalies have been marked, and their definitions explained:

- **Type 1 Anomaly:** When there is no power output above the cut in wind speed. This could be caused when the turbine is in its downtime. It is a contextual anomaly meaning it would be considered normal in a different context [\(Chandola et al., 2009\)](#).
- **Type 2 Anomaly:** When the power output is steady and continuous while the power output is less than the rated power. It is usually caused by curtailment. They are also contextual anomalies.
- **Type 3 Anomaly:** These are anomalies randomly scattered across the feature space. They may be generated by sensor malfunctions or signal processing noise. They are described as point anomalies, instances that do not conform to the rest of the data.

In the analysis, the presence of an unknown anomaly type was found when there is power output when wind speed is zero.

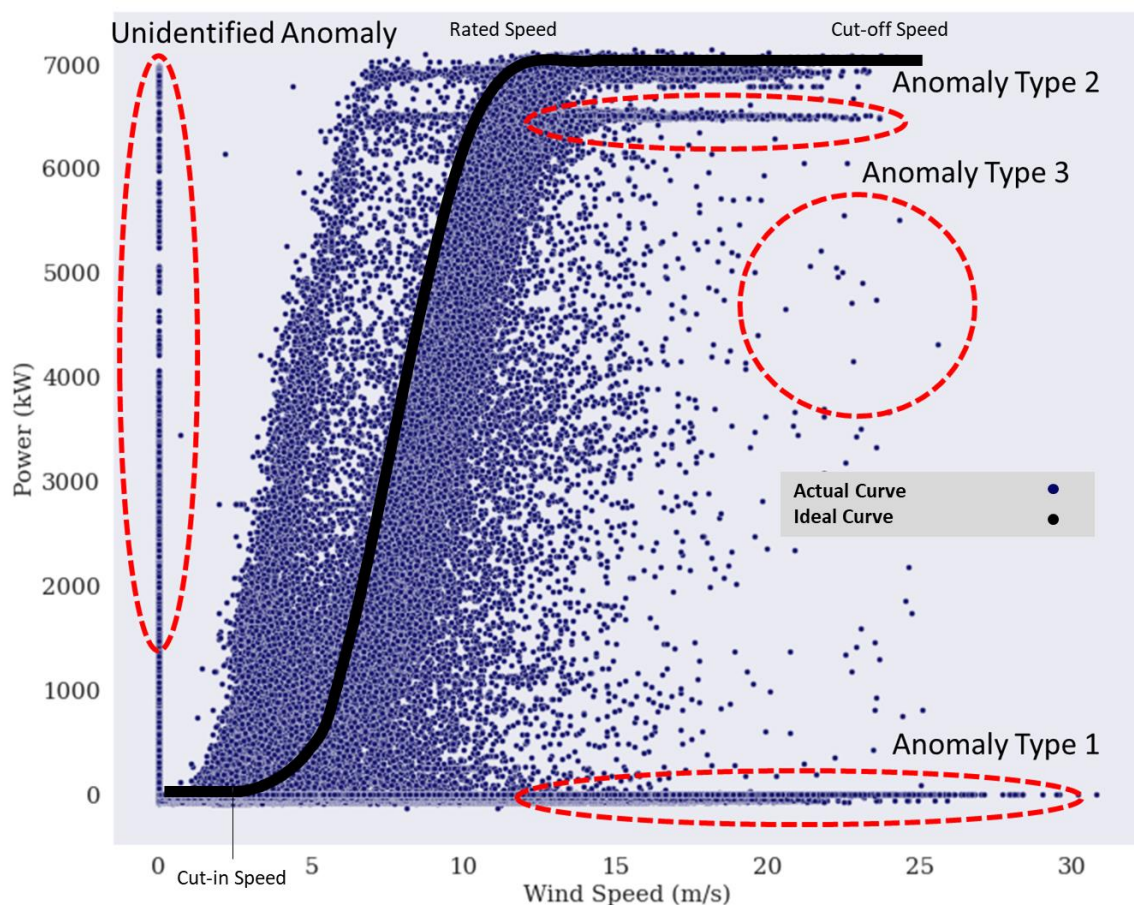


Fig 4 Turbine power curve

3.3. Pre-processing

3.3.1. Missing Values

From the descriptive statistics table, the presence of outliers is shown. Estimating values for missing pieces in a time series is relatively straightforward since it is appropriate to assume that missing observations would have values similar to the observations closest to them. In our case, we can confirm that entries close to each other are related based on the autocorrelation plot. Various techniques can be used for imputing missing values. Having said that, the best method is usually “Interpolation”. It estimates the missing value by finding an approximate value between a range of data points ([Earl & Nicholson, 2021](#)). It is better than other methods because it estimates the missing values based on past and future data points. For this project, Linear Interpolation was used. It is the simplest and fastest interpolation method. It simply draws a straight line through the two closest points, and the equation of a straight line can be used to determine the value of the point in-between.

3.3.2. Anomaly Detection and Removal

3.3.2.1. Temperature

From the descriptive statistics of the temperature column, it is clear that there are outliers. The mean temperature is -6.042120×10^{10} °C, and the minimum is -1.568601×10^{16} °C. These values are impossible as the coldest recorded temperature ever in Scotland was -27.2 °C ([Met Office, n.d.](#)). The median gives a value of 11.4C, showing clear evidence of outliers. After removing the clear outlier, the minimum value, the z-score of the temperature column, was measured, and values that had an absolute z-score value above five were removed and interpolated.

3.3.2.2. Power Curve

In the turbine’s power curve, three different types of anomalies were discovered, and uncategorised types were also discovered. These were cases where the power was negative or there was power output with zero windspeeds. These anomalies accounted for about 44% of the data and had to be removed using a filter. The filter also helped deal with type 1 anomalies by removing observations with no power output when the wind speed exceeded the cut-in speed (3 m/s). Finally, four standard outlier detection techniques were tested to see how well they identified the outliers in the dataset to remove the rest of the anomalies.

- **Local Outlier Factor:** “This is a density-based outlier detection method. It calculates the relative density measure outlier factor of each data relative to its surrounding points. This measure is then used to show the degree of the outlier in the data” ([Cheng et al., 2019](#)).
- **Isolation Forest:** This method isolates anomalies instead of profiling regular instances. It uses a tree structure. In the isolation process, normal instances are isolated at the deeper end of the tree, while anomalies are located close to the tree’s root ([Liu et al., 2008](#)).
- **Elliptical Envelope:** This method detects outliers in Gaussian distributions. It fits an ellipse to the central data points and ignores points outside the central mode; thus, the ignored points are classified as outliers ([Scikit-learn, b](#)).
- **Gaussian Mixture Model:** “It is a probabilistic model that assumes all data points are generated from a finite number of gaussian distributions and clusters” ([Scikit-learn, a](#)). In the case of our dataset, we chose 3 clusters since that gave the best Bayesian Information Criterion.

Fig 5 shows that the local outlier factor performs the best, which is why it is used. Out of all the models, it identified the most type 3 anomalies. However, it could not correctly identify the areas of curtailment as good as the isolation forest.

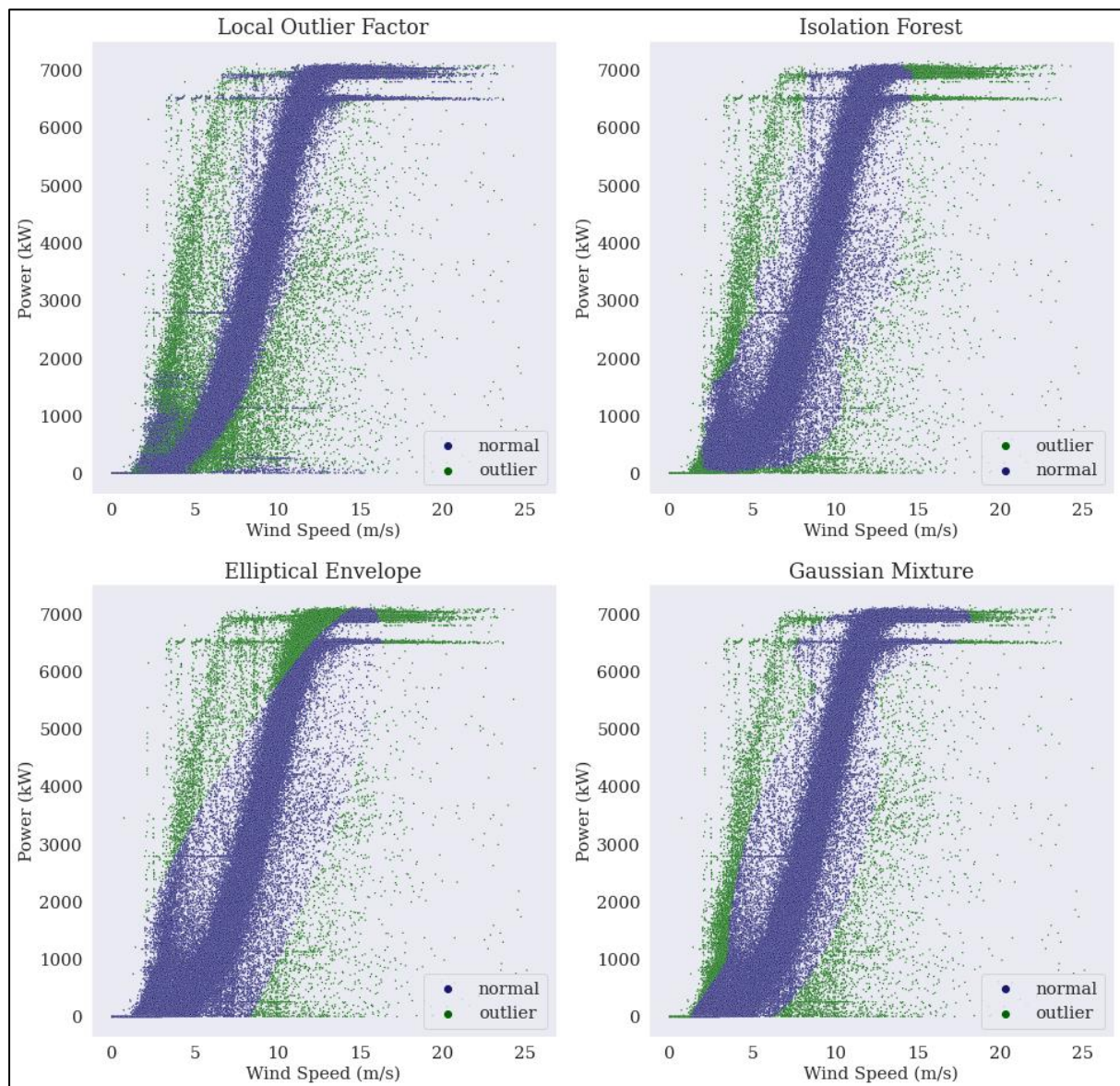


Fig 5 Shows the anomalies found after applying filter and four outlier detection techniques.

3.4 Feature Engineering

Feature engineering enhances the performance of a dataset on a machine learning model by transforming its feature space ([Nargesian et al., 2017](#)). Features were engineered to see if they could aid in improving the model.

U and V Components

The U and V components are commonly used in meteorology. They are vector representations of the wind speed and direction. They are a better way to show the direction of the wind than using degrees. The angles in degree might not work well due to the nature of angles, for example, 360 and 0 degrees are close to each other, but a machine learning model will view them as far apart since the numbers are very different. The u component represents the x-axis, while the v component represents the y-axis ([Virginia weather & climate data. 2014](#)). They are calculated using the equations:

$$u = ws * \cos \theta$$

$$v = ws * \sin \theta$$

ws = wind speed

θ = wind direction

Turbulence Intensity

Wind turbulence is de rapid fluctuations in wind speed. This factor can significantly impact the performance of the turbine ([Brower, 2012](#)). The formula for turbulence intensity is given below:

$$TI = \frac{\sigma}{v}$$

σ = standard deviation of wind speed

v = mean wind speed

Wind Shear

This is the rate of change of wind speed with height. It is expressed as alpha (α) , the power law exponent which is unitless ([Brower, 2012](#)). The influence of wind shear on the power production of wind turbines was analysed, and the findings were that when medium and high shear exponents were measured, the machine produced a lower power output than expected. That being said, more work is needed to test a broader range of wind turbines to confirm what was observed ([Honrubia et al., 2010](#)). The equation is given as:

$$\alpha = \frac{\log\left(\frac{v_2}{v_1}\right)}{\log\left(\frac{h_2}{h_1}\right)}$$

Where

v_2 = wind speed at height h_2 .

v_1 = wind speed at height h_1 .

Air Density

This is the mass per unit volume of the earth's atmosphere. This factor depends on temperature and pressure. The equation is given as:

$$\rho = \frac{P}{RT}$$

Where

P = Air Pressure

R = Specific gas constant (287.04 J/kg . K)

T = Temperature.

Normalisation

The features have different scales. This difference might cause the model to be biased toward larger-scale features. For this reason, normalisation is used to scale all features to a range between 0 and 1.

3.5 Machine Learning

The following algorithms were used for making predictions:

- **Random Forest:** A tree-based ensemble model. It combines multiple randomised decision trees and aggregates their predictions using the mean ([Biau & Scornet, 2016](#)). It used 200 decision trees and default parameters from sklearn.
- **XGBoost:** Another tree-based ensemble model that learns using a sequence of weak learners learning from their predecessors' mistakes. XGBoost is widely used and is known for achieving state-of-the-art results ([Chen & Guestrin, 2016](#)). The model used in this research had a series of 2000 weak learners, although it did not use up to that amount due to early stopping, meaning that training stops if the metrics did not improve after a certain number of training rounds.
- **Neural Networks** are a class of non-linear models made up of neurons, computing elements with weight and sometimes an activation function. These neurons are interconnected in complex ways and organised into layers, with inputs being fed into them, allowing them to create functions for the output ([Sarle, 1994](#)). Neural networks are good at making sense of non-linear data due to their activation functions ([Sharma et al., 2020](#)). For this project, two neural networks were used: one with five hidden layers and then a deeper one with eight hidden layers. The deeper model was implemented to see if it could improve performance as an experiment from Goodfellow et al. showed that deeper networks tend to generalise better ([2016](#)). Both models trained for 1000 epochs, although early stopping was used.

The dataset was divided into training, and testing sets in a ratio of 80:20. The features Wind Speed, Temperature, Pressure, U-Component, V-Component, Wind Shear, Turbulence Intensity and Air Density were the inputs. Power was the label.

4. Results

The table shows each machine learning model's metrics: the Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and R-Squared Score. The lower the value for RMSE and MAE, the better; ideally, they should be less than the mean of the label. The r-squared value should be as close to 1 as possible. XGBoost had the best metrics as it had the lowest RMSE and highest R^2 score on the testing set while achieving this in much less time than the other models. The Random Forest had the best testing MAE and altogether good testing metrics but overfitted the most as its training metrics were much better than testing. The neural networks had the worse metrics and longer training times, but on a positive note, they generalised better than both tree models with very similar training and testing metrics.

Table 2 Metrics of each machine learning model

Models	Train RMSE	Train MAE	Train R^2	Test RMSE	Test MAE	Test R^2	Time (s)	No estimators or epochs
Random Forest	160.97	84.22	0.99	430.65	227.70	0.96	198	200
XGBoost Model	326.21	197.01	0.98	424.94	237.71	0.97	29.9	174
5-Layer Neural Network	527.95	268.87	0.95	524.82	269.17	0.95	788	350
8-Layer Neural Network	489.06	253.99	0.95	485.11	254.92	0.95	483	119

Fig 6 plots the model's predictions against the actual values. It shows how well the predicted values fit the actual values of the dataset. The points should be as close to the dashed red line for a good fit as possible. Points outside the black lines show values far from the RMSE value, meaning their predictions were exceptionally far off.

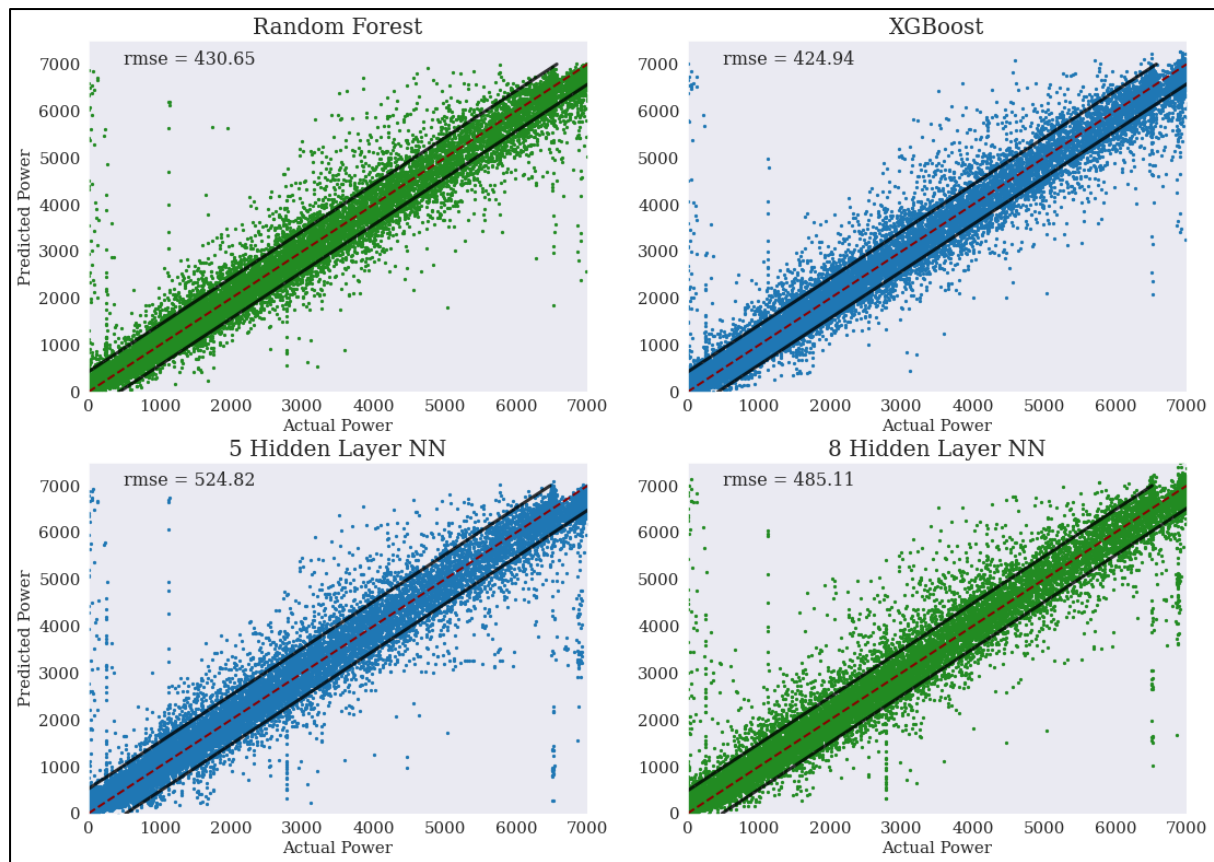


Fig 6 Predicted vs actual values plot

Feature importance assigns a score for the input features based on their impact on prediction. Above is the feature importance of the XGBoost. We can see that the wind speed is significantly the most important feature with about 93% score, followed by the wind shear, although it is relatively insignificant, as are the rest of the values.

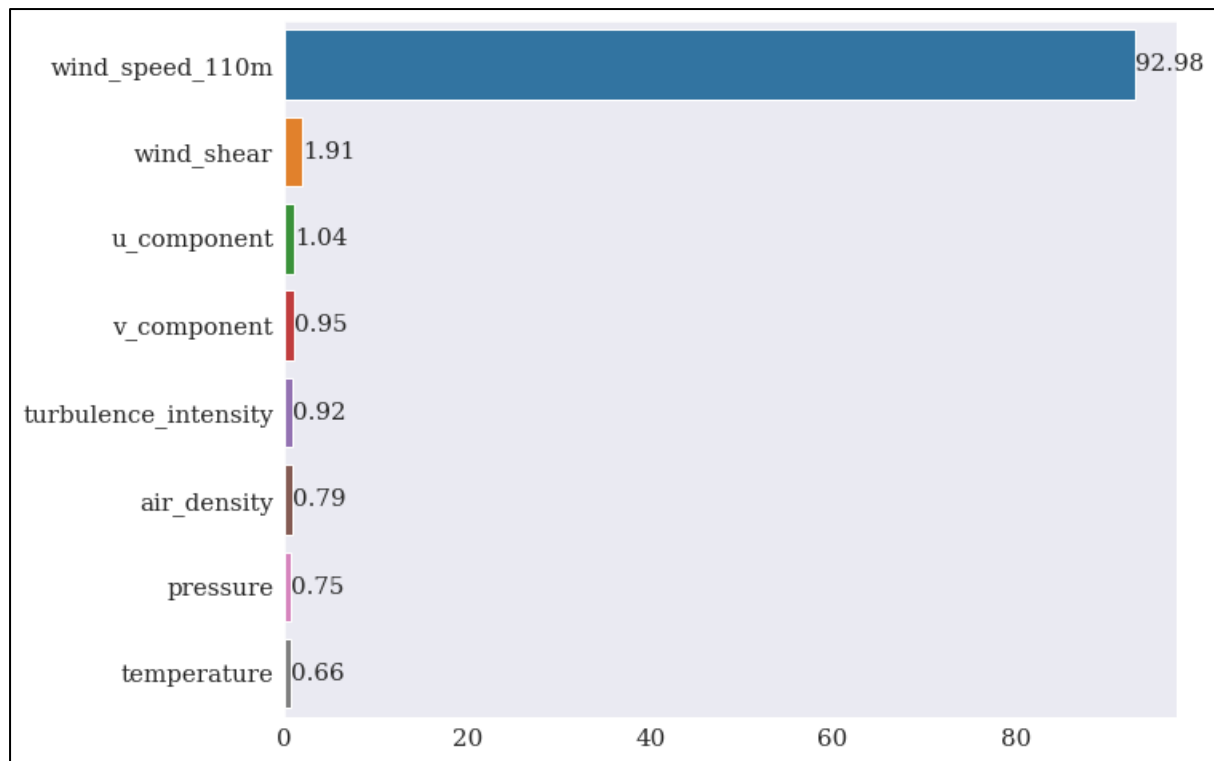


Fig 7 Feature importance of input features from XGBoost

5. Discussion

Both tree-based methods (Random Forest and XGBoost) had the best metrics. This is expected as tree-based models are superior in tabular datasets, even outperforming neural networks, which are dominant with structured data where the relative position is vital, like in images, audios and text ([Xu et al., 2021](#); [Grinsztajn et al., 2022](#)). Although the neural networks performed worse than tree-based, significantly more architectural options can be tested. The 8-layer network had slightly better metrics than the 5-layer network. Theoretically, an even deeper architecture could improve the metrics, although it will increase training time and overfit the data.

When compared with previous work, it is hard to make a comparison since others used different turbine datasets with differing rated power outputs, although based on the R^2 compared with others like ([Clifton et al., 2013](#)) and ([Alkesaiberi et al., 2022](#)), who had values as high as 0.99 and 0.97 we can see that our models performed adequately.

The feature importance plot in **fig 7** shows that wind speed is the most essential feature. This is expected as the power generated is a direct effect of wind speed, but looking at the factors, they have negligible impact in predicting the power. Therefore, it raises the question of whether there is a need to use the other features in forecasting, as their presence is highly negligible in the models.

From the Actual vs Predicted plot in **fig 6**, we can see that although predicted values are not precisely in line with the actual values, they still follow the trends. However, there are still points incredibly far away from the line. The RMSE for all models is significantly higher than the MAE because of these variations, as RMSE is sensitive to larger errors. We can see how far some points are from the black lines. The variations might be because the noise was still present even after using the anomaly detection techniques and filters. Although in most real-world datasets, there is always noise, the presence of noise might help the model generalise better. Then again, their presence can also lead to poor prediction results ([Gupta & Gupta, 2019](#)).

6. Conclusion

In conclusion, the models performed well with decent metrics, although there is still a chance for improvement using different approaches. From the Autocorrelation plot in **fig 3**, we can see high autocorrelation in both power and wind speed. This means a time series model might be able to produce better metrics. A multivariate model that looked at previous wind speed and power values to make predictions could work well. A wide array of neural network architectures can deal with sequential data like Recurrent Neural Networks, Convolutional Neural Networks and Transformers. Although in this project, that approach could not be taken because of the dataset. About 44% of the power output values were negative, and with no viable option for imputing, they had to be removed, causing the data to lose its sequential nature.

A time-series approach could be implemented in future experiments with a better dataset with fewer anomalies. However, this option may not always be viable as data from SCADA systems usually have significant error values. Taking sequential subsections of the data with less noise could help mitigate the issues of using a time series model, although this would still lead to data loss.

7. References

Virginia weather & climate data. (2014) Available online:

<http://colaweb.gmu.edu/dev/clim301/lectures/wind/wind-uv> [Accessed 17/08/ 2022].

Alkesaiberi, A., Harrou, F. & Sun, Y. (2022) Efficient wind power prediction using machine learning methods: A comparative study. *Energies (Basel)*, 15 (7), 2327.

Anderson, A., Semmelroth, D. & Dummies, C. (2015) *Statistics for big data for dummies*. New York: Wiley.

Biau, G. & Scornet, E. (2016) A random forest guided tour. *Test (Madrid, Spain)*, 25 (2), 197-227.

Brower, M. C. (2012) *Wind resource assessment : A practical guide to developing a wind project*, 1. Aufl. edition. Somerset: John Wiley & Sons, Inc.

Chandola, V., Banerjee, A. & Kumar, V. (2009) Anomaly detection. *ACM Computing Surveys*, 41 (3), 1-58.

Chen, T. & Guestrin, C. (2016) XGBoost. Ithaca: ACM.

Cheng, Z., Zou, C. & Dong, J. (2019) Outlier detection using isolation forest and local outlier factor. ACM.

Clifton, A., Kilcher, L., Lundquist, J. K. & Fleming, P. (2013) Using machine learning to predict wind turbine power output. *Environmental Research Letters*, 8 (2), 1-8.

Earl, R. & Nicholson, J. (2021) *Theæ concise oxford dictionary of mathematics*, Sixth edition edition. Oxford: Oxford University Press.

European Wind Energy Association (2009) *Wind energy - the facts*. London: Routledge.

Goodfellow, I., Bengio, Y. & Courville, A. (2016) *Deep learning*. Cambridge, Massachusetts ; London, England: The MIT Press.

Grinsztajn, L., Oyallon, E. & Varoquaux, G. (2022) Why do tree-based models still outperform deep learning on tabular data? .

Gupta, S. & Gupta, A. (2019) Dealing with noise problem in machine learning data-sets: A systematic review. *Procedia Computer Science*, 161 466-474.

Guthrie, W. F. (2020) NIST/SEMATECH e-handbook of statistical methods (NIST handbook 151). National Institute of Standards and Technology.

Hanifi, S., Liu, X., Lin, Z. & Lotfian, S. (2020) A critical review of wind power forecasting Methods—Past, present and future. *Energies (Basel)*, 13 (15), 3764.

Honrubia, A., Vingueras-Rodriguez, A., Lazaro, E. G., Rodriguez, D., Mejias, M. & Lainez, I. (2010) The influence of wind shear in wind turbine power estimation. *Measurements*, 2 9.

Khan, M., Naeem, M. R., Al-Ammar, E. A., Ko, W., Vettikalladi, H. & Ahmad, I. (2022) Power forecasting of regional wind farms via variational auto-encoder and deep hybrid transfer learning. *Electronics (Basel)*, 11 (2), 206.

Lew, D., Milligan, M., Jordan, G. & Piwko, R. (2011) Value of wind power forecasting.

Lin, Z., Liu, X. & Collu, M. (2020) Wind power prediction based on high-frequency SCADA data along with isolation forest and deep learning neural networks. *International Journal of Electrical Power & Energy Systems*, 118 105835.

Liu, F. T., Ting, K. M. & Zhou, Z. (2008) Isolation forest. *2008 eighth IEEE international conference on data mining*. IEEE.

Lydia, M., Kumar, S. S., Selvakumar, A. I. & Prem Kumar, G. E. (2014) A comprehensive review on wind turbine power curve modeling techniques. *Renewable & Sustainable Energy Reviews*, 30 452-460.

Met Office (n.d.) *UK climate extremes*. Available online: <https://www.metoffice.gov.uk/research/climate/maps-and-data/uk-climate-extremes#?tab=climateExtremes> [Accessed 16/08/ 2022].

Ministry of Defence. (2021) AIR DEFENCE AND OFFSHORE WIND - WORKING TOGETHER TOWARDS NET ZERO, *States News Service*, [Online]. Available at: <https://www.gov.uk/government/publications/air-defence-and-offshore-wind-working-together-towards-net-zero/air-defence-and-offshore-wind-working-together-towards-net-zero> (Accessed 16/08/2022).

Morrison, R., Liu, X. & Lin, Z. (2022) Anomaly detection in wind turbine SCADA data for power curve cleaning. *Renewable Energy*, 184 473-486.

Nargesian, F., Samulowitz, H., Khurana, U., Khalil, E. B. & Turaga, D. S. (2017) Learning feature engineering for classification. *Ijcai*.

ORE Catapult (n.d.) *Levenmouth demonstration turbine*. Available online: <https://pod.ore.catapult.org.uk/source/levenmouth-demonstration-turbine> [Accessed 16/08/ 2022].

Ørsted, U. K. (n.d.) *Offshore wind*. Available online: https://orsted.co.uk/energy-solutions/offshore-wind?gclid=CjwKCAjwoMSWBhAdEiwAVJ2ndmqEWupYQcEPWZMVsmj7UOh-Mi9Zkfeb-4ehH0_x9Suu0yBgpmSYLRoC-sQAvD_BwE [Accessed 16/08/ 2022].

Sarle, W. S. (1994) Neural networks and statistical models.

Saroha, S., Aggarwal, S. K. , Rana, P. (2021) *Forecasting in mathematics - recent advances, new perspectives and applications*. London: IntechOpen.

Scikit-learn (a) *Gaussian mixture models*. Available online: <https://scikit-learn.org/stable/modules/mixture.html#:~:text=A%20Gaussian%20mixture%20model%20is,Gaussian%20distributions%20with%20unknown%20parameters.> [Accessed 16/08/ 2022].

Scikit-learn (b) *Novelty and outlier detection*. Available online: https://scikit-learn.org/stable/modules/outlier_detection.html#outlier-detection [Accessed 16/08/ 2022].

Sharma, S., Sharma, S. & Athaiya, A. (2020) Activation functions in neural networks. *International Journal of Engineering Applied Sciences and Technology*, 4 (12), 310-316.

Skiena, S. S. (2017) *The data science design manual*. Cham: Springer International Publishing AG.

Sohoni, V., Gupta, S. C. & Nema, R. K. (2016) A critical review on wind turbine power curve modelling techniques and their applications in wind based energy systems. *Journal of Energy*, 1-18.

UK Government. (2021) Net zero strategy: Build back greener. Dandy Booksellers.

Xu, H., Kinfu, K. A., LeVine, W., Panda, S., Dey, J., Ainsworth, M., Peng, Y., Kusmanov, M., Engert, F., White, C. M., Vogelstein, J. T. & Priebe, C. E. (2021) When are deep networks really better than decision forests at small sample sizes, and how? .

Younas Zubair (2019) *Why grid active power P comes -ive and how i can overcome it?* Available online: <https://www.researchgate.net/post/Why-grid-active-power-P-comes-ive-and-how-i-can-overcome-it> .